PYTHON DEVELOPMENT INTERNSHIP

WEEK 6 PROJECT REPORT

STUDENT RECORD MANAGEMENT SYSTEM

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Internship Domain: Python Development

Organization: InternNova

Week: Week 6 of 6

Project Type: Python Console Application

1. Introduction

The Student Record Management System is a Python-based console application developed to manage student information in an organized way. The application allows users to add, view, search, update, and delete student records.

The project uses Object-Oriented Programming (OOP) concepts to organize the code and JSON file handling to store student information. It is designed to make basic student record management simple and easy to understand.

2. Problem Statement

Managing student information manually can be time-consuming and may lead to mistakes or difficulties in updating records. A simple computerized system can help organize student details and make common operations easier.

The purpose of this project is to develop a Python application that provides basic student record management features and stores information in a JSON file.

3. Objectives

The main objectives of this project are:

- To develop a menu-driven student record management application.
- To add, view, search, update, and delete student records.
- To store student information using JSON file handling.
- To understand and implement Object-Oriented Programming concepts.
- To validate student marks and calculate grades.
- To calculate the average marks of student records.
- To improve practical Python programming and problem-solving skills.

4. Features of the Project

The application is designed with the following features:

1.Add Studends: Allows the user to enter student details, including student ID, name, age, course, and marks.

2. View Students: Displays the student records stored in the system.

3. Search Student: Searches for a student using the student ID.

4. Update Student: Allows the user to modify the details of an existing student.

5. Delete Student: Removes a student record using the student ID.

6. Grade Calculation: Calculates a grade based on the marks entered.

7. Average Marks: Calculates the average marks of the available student records.

8. JSON File Storage: Uses a JSON file to save and load student information.

9. Input Validation: Checks that marks are within the permitted range.

Note: These are the intended project features. Each feature should be tested before the final submission.

5. Tools and Technologies Used

- Programming Language: Python 3
- Code Editor: Visual Studio Code
- Data Storage: JSON
- Python Module: json
- Version Control: Git
- Code Hosting: GitHub

6. Concepts Used

6.1 Classes and Objects

Classes are used to organize data and functions. Objects are created from classes to represent students and manage student records.

6.2 Inheritance

The Student class inherits from the Person class. This allows the student class to reuse relevant attributes and behaviour from the parent class.

6.3 Encapsulation

The marks property is used to control how marks are assigned and validated. This helps prevent invalid marks from being accepted through the property.

6.4 Method Overriding

The Student class provides its own show_details() method. This demonstrates method overriding in Python.

6.5 File Handling

File handling is used to read and write student information. The application uses a JSON file to preserve records between program runs.

6.6 Exception Handling

Exception handling helps the program deal with situations such as a missing data file or invalid JSON data without immediately terminating.

6.7 Conditional Statements and Functions

Conditional statements are used to process menu choices and calculate grades. Functions and methods divide the program into smaller, manageable parts.

7. Working Procedure

The application follows these steps:

1. The user runs the main.py file.
2. The program displays a menu containing the available operations.
3. The user selects an operation.
4. The user enters the required student information or student ID.
5. The program processes the selected operation and displays the result.
6. Student records are saved in the students.json file when the application saves the data.
7. The user can select the exit option to close the application.

8. Project Files

The project contains the following files:

- main.py: Contains the Python classes, functions, application logic, and menu.
- students.json: Stores student records in JSON format. It can begin with an empty list ([]).
- README.md: Contains the project description, features, and instructions for running the application.
- screenshots/: A folder intended to contain screenshots of the running application.

9. Testing and Results

The application has been launched successfully, and the main menu has been displayed. The Add Student operation has also been tried successfully.

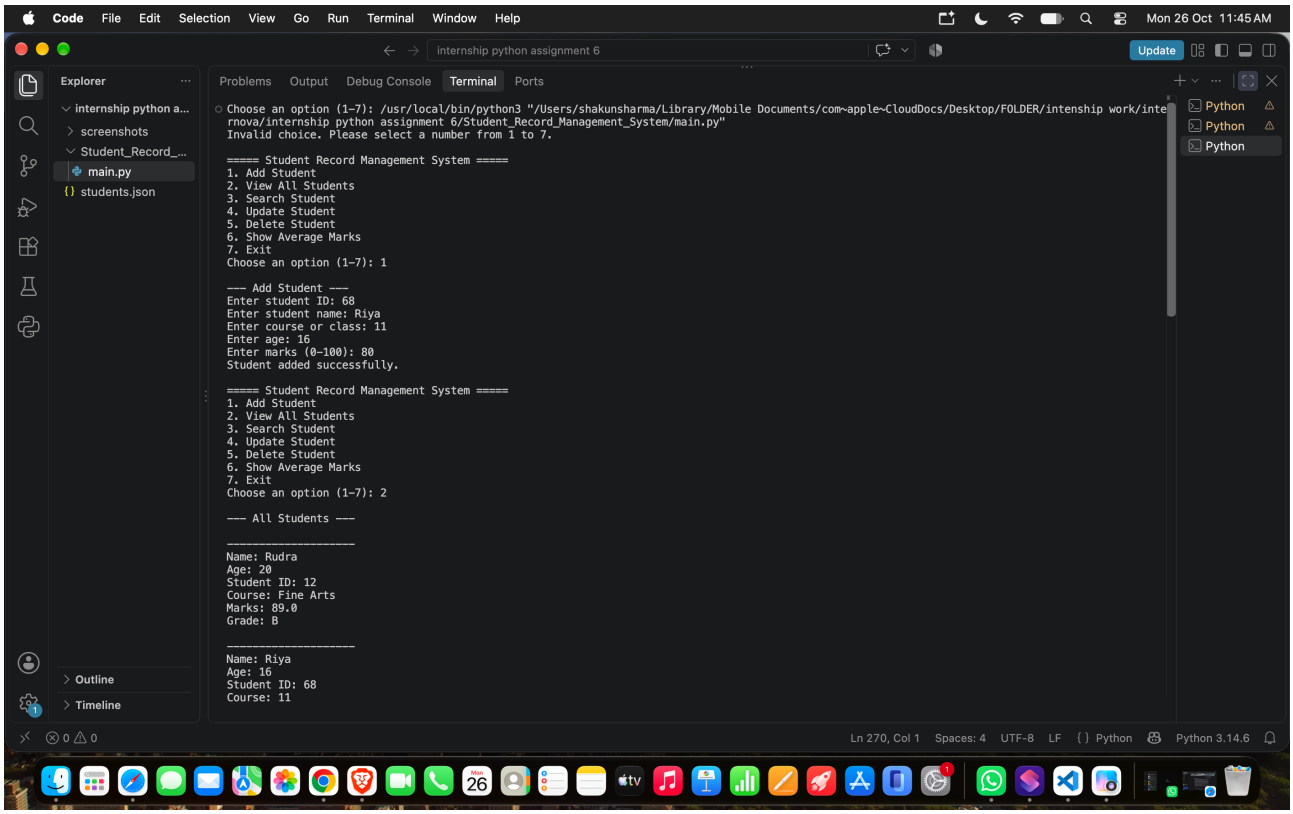
The remaining operations should be tested before final submission to confirm that they work as expected.

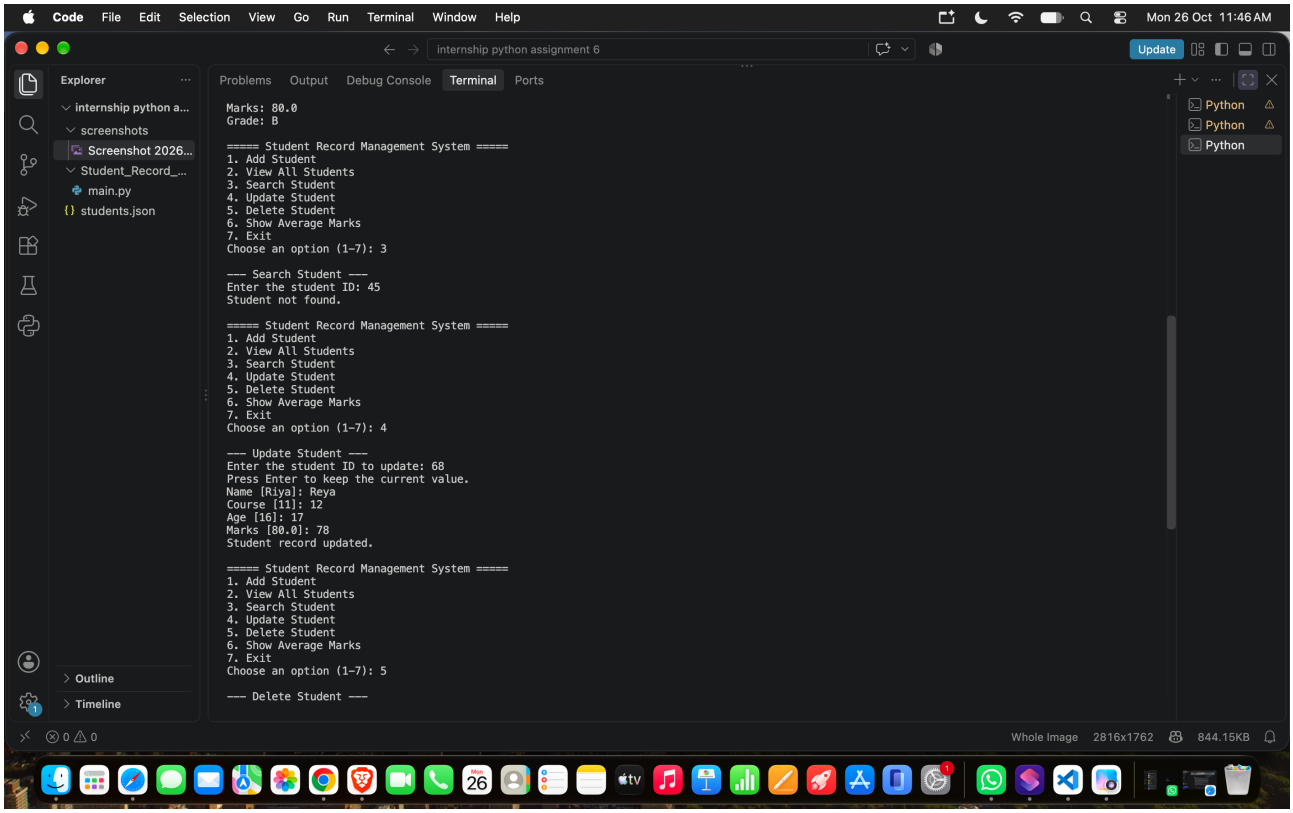
The following test cases are planned:

1. Adding a student with valid details.
2. Viewing saved student records.
3. Searching for an existing and a non-existing student ID.
4. Updating an existing student record.
5. Deleting a student record.
6. Checking grade and average-mark calculations.
7. Restarting the program to check whether saved data loads correctly.
8. Entering invalid marks to check input validation.

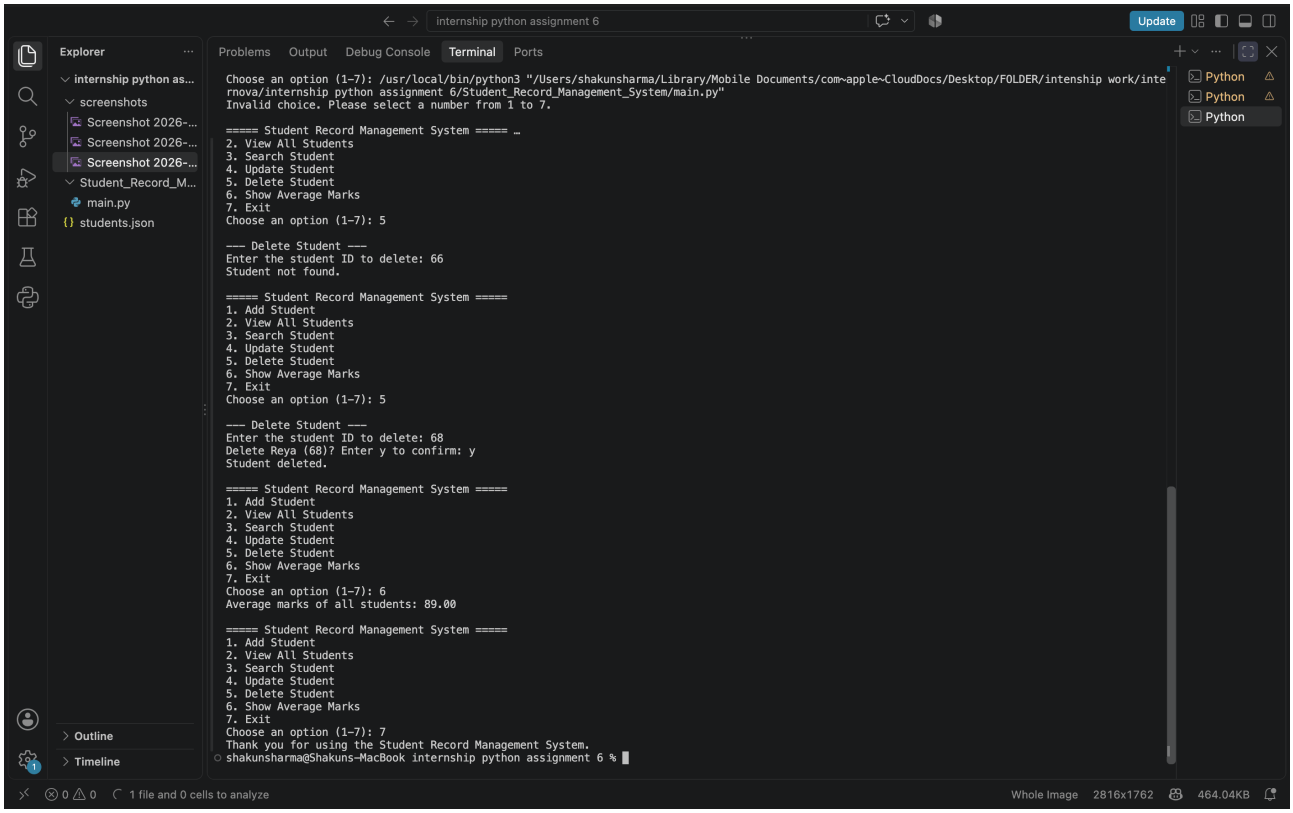
Screenshots

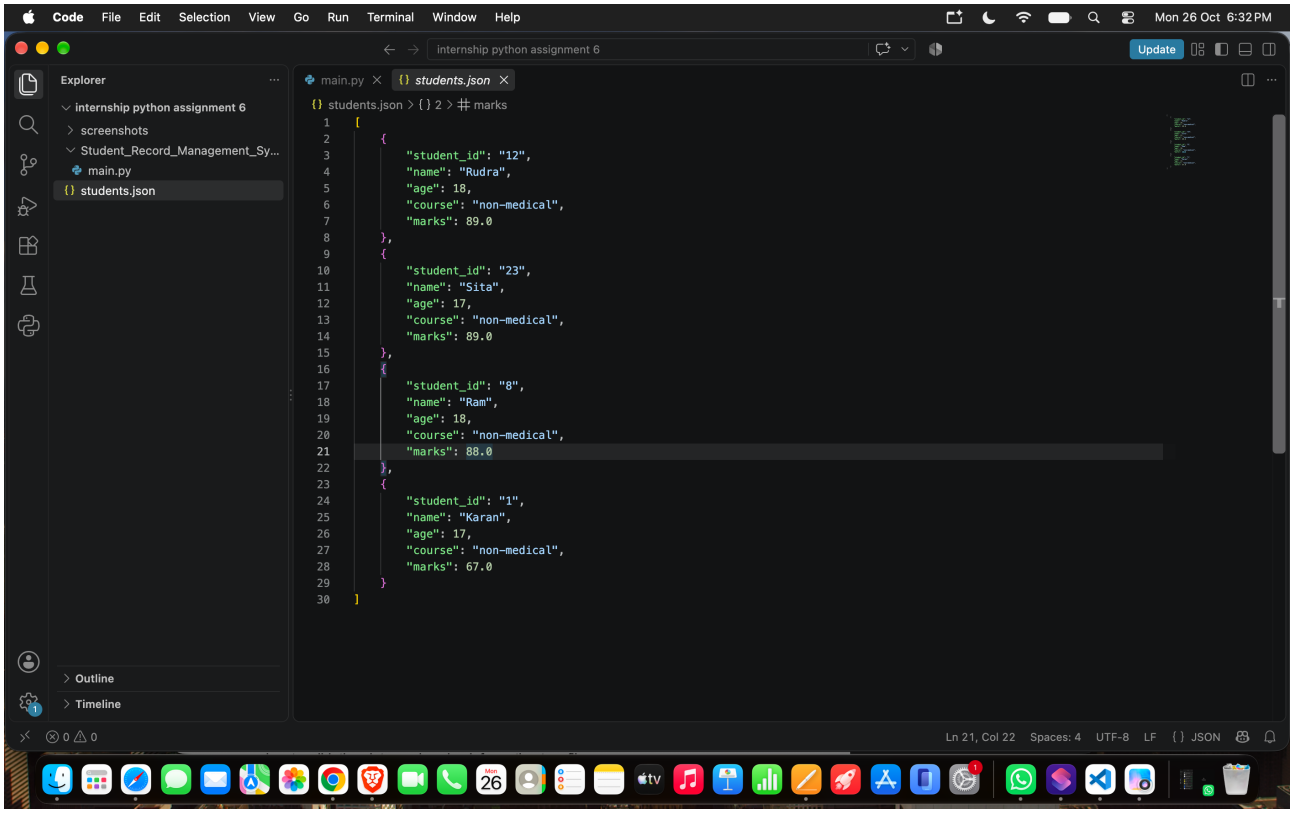
Insert your actual screenshots under the following headings:




Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 11:46 AM
internship python assignment 6
Update
Explorer
internship python a...
screenshots
Screenshot 2026...
Student_Record_...
main.py
students.json
Problems Output Debug Console Terminal Ports
Python
Python
Python
Marks: 80.0
Grade: B
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 3
--- Search Student ---
Enter the student ID: 45
Student not found.
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 4
--- Update Student ---
Enter the student ID to update: 68
Press Enter to keep the current value.
Name [Riya]: Reya
Course [11]: 12
Age [16]: 17
Marks [80.0]: 78
Student record updated.
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 5
--- Delete Student ---
Outline
Timeline
Whole Image 2816x1762 844.15KB


Screenshot 2026-10-26 at 11.46.18 AM.png — internship python assignment 6
Explorer
internship python as...
screenshots
Screenshot 2026-...
Screenshot 2026-...
Student_Record_M...
main.py
students.json
Problems Output Debug Console Terminal Ports
Python
Python
Python
Course [11]: 12
Age [16]: 17
Marks [80.0]: 78
Student record updated.
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 5
--- Delete Student ---
Enter the student ID to delete: 66
Student not found.
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 5
--- Delete Student ---
Enter the student ID to delete: 68
Delete Reya (68)? Enter y to confirm: y
Student deleted.
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7): 6
Average marks of all students: 89.00
===== Student Record Management System =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1-7):
Outline
Timeline
Whole Image 2816x1762 843.60KB


internship python assignment 6
Update
Explorer
internship python as...
screenshots
Screenshot 2026-...
Screenshot 2026-...
Screenshot 2026-...
Student_Record_M...
main.py
students.json
Problems
Output
Debug Console
Terminal
Ports
Choose an option (1–7): /usr/local/bin/python3 "/Users/shakunsharma/Library/Mobile Documents/com~apple~CloudDocs/Desktop/FOLDER/intenship work/internova/internship python assignment 6/Student_Record_Management_System/main.py"
Invalid choice. Please select a number from 1 to 7.
===== Student Record Management System =====
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Average Marks
7. Exit
Choose an option (1–7): 5
--- Delete Student ---
Enter the student ID to delete: 66
Student not found.
Choose an option (1–7): 5
--- Delete Student ---
Enter the student ID to delete: 68
Delete Reya (68)? Enter y to confirm: y
Student deleted.
Choose an option (1–7): 6
Average marks of all students: 89.00
Choose an option (1–7): 7
Thank you for using the Student Record Management System.
shakunsharma@Shakuns-MacBook internship python assignment 6 %
Python
Python
Python
Outline
Timeline
1 file and 0 cells to analyze
Whole Image
2816x1762
464.04KB


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 6:32 PM
internship python assignment 6
Update
Explorer
internship python assignment 6
screenshots
Student_Record_Management_Sy...
main.py
students.json
main.py
students.json
{} students.json > { } 2 > # marks
[
{
"student_id": "12",
"name": "Rudra",
"age": 18,
"course": "non-medical",
"marks": 89.0
},
{
"student_id": "23",
"name": "Sita",
"age": 17,
"course": "non-medical",
"marks": 89.0
},
{
"student_id": "8",
"name": "Ram",
"age": 18,
"course": "non-medical",
"marks": 88.0
},
{
"student_id": "1",
"name": "Karan",
"age": 17,
"course": "non-medical",
"marks": 67.0
}
]
Outline
Timeline
Ln 21, Col 22
Spaces: 4
UTF-8
LF
{} JSON

10. Challenges Faced

One challenge during the initial setup was running the Python file from the correct directory. Since main.py is inside the project folder, the program must be started from that folder or by specifying the correct file path.

This helped me understand the importance of project structure and the current working directory when running Python programs.

The project also provides practical experience in organizing code with classes, handling user input, validating data, and saving information to a file.

11. Future Improvements

The project can be improved in the future by:

- Adding stronger validation for student names, ages, and courses.
- Preventing duplicate student IDs.
- Improving error handling for damaged or incorrectly formatted JSON files.
- Adding sorting and filtering options.
- Writing automated tests for the application.
- Developing a graphical user interface or a web-based version.
- Improving the design and user experience.

12. Conclusion

The Student Record Management System is a practical Python project that demonstrates the use of fundamental programming concepts and Object-Oriented Programming. It is designed to simplify common student record management operations through a menu-driven interface.

Through this project, I have gained practical experience with classes, inheritance, method overriding, input validation, exception handling, and JSON file storage. After testing all features, adding screenshots, and uploading the project to GitHub, the project will be ready for final submission.

13. GitHub Repository

Project Name: Student Record Management System

GitHub Repository URL: [Paste your repository link here]

The repository should contain the project source code, README file, and any other required supporting files before submission.